

# Constrained sphere packing and the interpolation problem for positive definite functions

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Joint work with Sujit Sakharam Damase, IISc, Bangalore

[illegible]

2 / 29

# Acknowledgments

We would like to thank:

- The Indian Institute of Science, Bangalore
- The Indian Statistical Institute, Bangalore
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Thanks to Jeff Remmel (1948-2017)  
for teaching me representation theory.



Permanently building the network in all areas with an understanding of the transient nature of political entities is key to a continuing human role in mathematical progress.

# Sphere packing

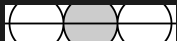
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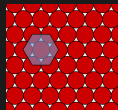
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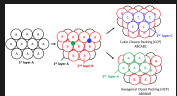
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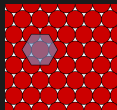
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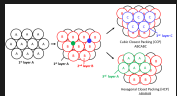
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For no other cases is the optimal density known exactly, although the optimal packings do not come from lattices.



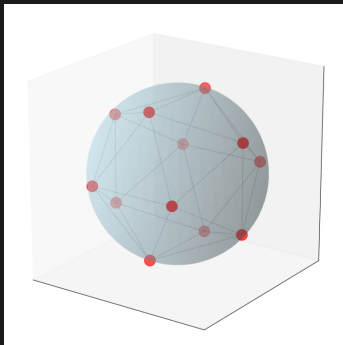
# Spherical codes

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A **spherical code** is a set of points in the sphere in  $n$  real dimensional space. We say it has **minimum angle** to be the smallest  $\theta$  such that every pair of distinct points in the code makes an angle of at least  $\theta$  with respect to the origin.

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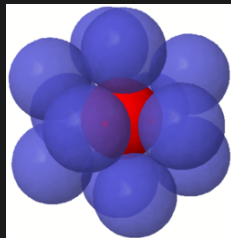
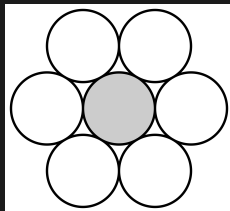
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The kissing number is equal to the maximal size of a spherical code with minimum angle  $\pi/3$ .

# Delsarte's method

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**Observation:** Let  $\pi/2 < \theta < \pi$ . Any spherical code with minimum angle  $\theta$  has cardinality less than

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**Strategy:** Given  $n \in \mathbb{N}$  and  $0 < \theta < \pi/2$  find the maximum  $\tilde{\theta} > \pi/2$  such that there is a map from the sphere in  $n$ -dimensional space to the sphere in a larger dimensional space such that every spherical code with minimum angle  $\theta$  is mapped to a code with minimum angle at least  $\tilde{\theta}$ .

# Positive definite functions

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Let  $G$  be group, let  $H$  be subgroup of  $G$ . A (completely) positive definite function

$$f : G/H \rightarrow \mathbb{R} \text{ (resp. self-adjoint operators)}$$

satisfies:  $[f(g_i^* g_j)]_{i,j}$  is positive semidefinite for all  $g_1, \dots, g_d \in G$ .

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Given a positive definite function  $f : G/H \rightarrow \mathbb{R}$  it has a **Stinespring factorization**

$$f(g) = v^* \pi(g) v$$

where  $\pi$  is a representation of  $G$  and  $v$  is a vector which is invariant for  $\pi|_H$ . If  $f$  is continuous and  $G$  is a compact group, the Peter-Weyl decomposition gives this is a direct sum of irreducibles. (convex combination)

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The homomorphic analogue of **Tomiyama's theorem** gives that  $f$  is in fact bi-invariant,

$$f(h_1 g h_2) = f(g), \forall h_1, h_2 \in H, g \in G.$$

# Schoenberg's theorem on positive definite functions

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Let  $G = SO(n)$ ,  $H = 1 \oplus SO(n-1)$ ,  $S^{n-1} \cong G/H$ . By bi-invariance we have that  $f(\gamma)$  depends only on the 1-1 entry of  $\gamma$ .



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If  $f$  is positive definite and continuous,  $f(g) = \sum_{i=1}^{\infty} a_i C_i^{(n)}(\gamma_{11})$  where  $a_i \geq 0$  and the  $C_i^{(n)}$  are Gegenbauer polynomials, a natural basis for bi-invariant functions. (We will normalize to have  $C_i^{(n)}(1) = 1$ )  
(Schoenberg '42, Bochner '41)

# Operations on positive definite functions

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Given  $C_i^{(n)}(\gamma_{11}) = v_i^* \pi_i(\gamma) v_i$ , we see that

$$C_i^{(n)} C_j^{(n)} = \sum_{k=0}^{i+j} a_k C_k^{(n)}$$

where  $\sum a_k = 1, a_k \geq 0$  as

$$C_i^{(n)}(\gamma_{11}) C_j^{(n)}(\gamma_{11}) = (v_i \otimes v_j)^* (\pi_i \otimes \pi_j)(\gamma) (v_i \otimes v_j)^*$$

and  $\pi_i \otimes \pi_j$  may again be broken into irreducibles.

# Kernel embedding

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## Theorem (Aronszajn's theorem)

*Let  $X$  be a set. Let  $A : X \times X \rightarrow \mathbb{C}$  such that  $[A(x_i, x_j)]_{i,j}$  is positive semidefinite for arbitrary finite collections  $x_1, \dots, x_d$ .*

*There exists a Hilbert space  $\mathcal{H}$  and  $\iota_x : X \rightarrow \mathcal{H}$  such that  $\langle \iota_x, \iota_y \rangle = A(x, y)$ .*

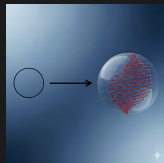
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Thus a positive definite function  $f : [-1, 1] \rightarrow \mathbb{R}$  such that  $f(1) = 1$  induces an embedding via the kernel  $f(\langle x, y \rangle)$  which lands in the sphere of the Hilbert space as  $f(\langle x, x \rangle) = 1$ .



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## Theorem (Delsarte '72)

*Let  $0 < \theta < \pi$ . Let  $g : [-1, 1] \rightarrow \mathbb{R}$  be a positive definite function on the sphere in  $n$  dimensions such that  $g(1) = 1$  and  $g(x) \leq 0$  if  $x \leq \cos \theta$ .*

*Let  $\bar{g}$  the constant coefficient of  $g$  in its Gegenbauer expansion. (That is the average value of  $g(\langle x, y \rangle)$  over  $x, y$  in the unit sphere. Let  $\mathcal{C}$  be spherical code with minimal angle  $\theta$ . Then,*

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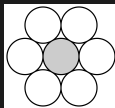
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Linear programming methods: Delsarte-Goethals-Seidel, '77 Kabitiyansky-Levenshtein '78.

# Kissing number in $\mathbb{R}^2$

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The polynomial

$$p(t) = \frac{4}{9}(t - 1/2)(t + 1/2)^2(1 + t)$$

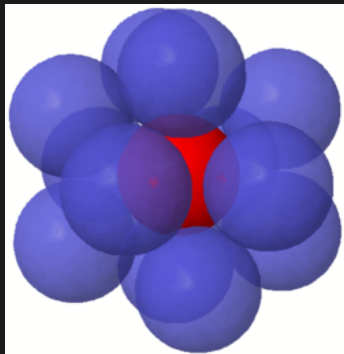
satisfies

$$p(x) = \frac{1}{6}C_0^{(2)}(x) + \frac{1}{3}C_1^{(2)}(x) + \frac{5}{18}C_2^{(2)}(x) + \frac{1}{6}C_3^{(2)}(x) + \frac{1}{18}C_4^{(2)}(x)$$

$$\text{i.e. } p(\cos \theta) = \frac{1}{6} + \frac{1}{3} \cos \theta + \frac{5}{18} \cos 2\theta + \frac{1}{6} \cos 3\theta + \frac{1}{18} \cos 4\theta$$

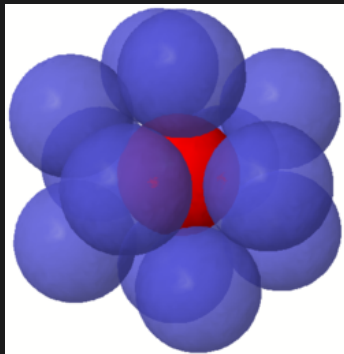
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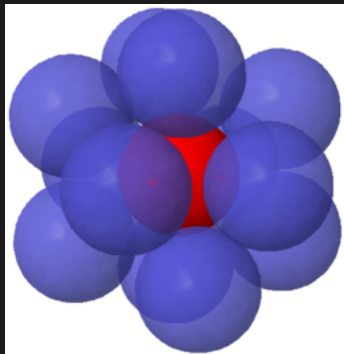
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Sharp

$n = 2$  kissing number

$n = 8, 24$  kissing number

Sloane, Odlyzko, 79

Not Sharp

$n = 3, 4$  kissing number

Musin

$X = \{-1, 1/2, -1/2, 1\}, n = 10$

Damase, P. ( $\frac{1}{\bar{g}} = 46.$ )

# Cohn-Elkies method

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## Theorem (Cohn-Elkies, '03)

*Suppose  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is an admissible function, is not identically zero, and satisfies the following two conditions:*

- 1  $f(x) \leq 0$  for  $|x| \geq 1$ , and
- 2  $\hat{f}(t) \geq 0$  for all  $t$ .

*Then the center density of  $n$ -dimensional sphere packings is bounded above by*

$$\frac{f(0)}{2^n \hat{f}(0)}.$$

# Constrained angle codes

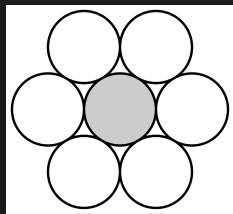
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For example, the spherical code given by six equispaced points on the circle has cosines in  $\{0, -1, 1/2, -1/2\}$ .

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When is there a positive definite function on the sphere in  $n$ -dimensions  $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$  such that  $\tilde{f}|_X = f$ ?

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c.f. Krein, Jorgensen, etc. on similar but different interpolation problems.



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Theorem (Damase-P.)

$SO(\infty)/1 \oplus SO(\infty)$  and  $SO(2)/\{1\}$  are not pathological.

# Continuous extension problem

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Let  $X$  be a finite subset of  $[-1, 1]$ . Let  $f : X \rightarrow \mathbb{R}$ . When is there a continuous positive definite function on the sphere in  $n$ -dimensions  $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$  such that  $\tilde{f}|_X = f$ ?

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Theorem (Damase, P.)

*Let  $f$  correspond to a solvable continuous extension problem. There is a polynomial solution  $\tilde{f}$ .*



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## Theorem (Nonantipodal sharpness)

*Let  $-1 \notin X$ . If  $X$  is Delsarte sharp, then there exists a positive definite polynomial  $g$  with  $\frac{1}{g} = |C|$ .*

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*If  $X = -X$  is Delsarte sharp, then any even function  $h$  such that  $h - 2\bar{g}$  is positive definite,  $h|_{X \setminus \{\pm 1\}} \leq 0$  and  $h(1) = 1$  is a polynomial.*

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Viazovska et al. functions for Cohn-Elkies are modular functions. (Low algebraic complexity.)

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- Consider  $X = \{-1, 1/2, -1/2, 1\}$ ,  $n = 10$ . There is no corresponding constrained angle code, but we have  $\frac{1}{g} = 46$  is an integer. Many such examples exist of **Delsarte hallucinations**.

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- On the other hand, the example  $X = \{1/3, -1/3, 1\}$  has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.
- Given an  $X$ , determining if there exists a subset of the  $n$ -sphere with cosines in  $X$  of some size  $k$  is a difficult combinatorial matrix completion problem.

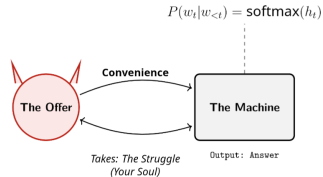


# The future is now

Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.



—Sir Michael Atiyah, 2002

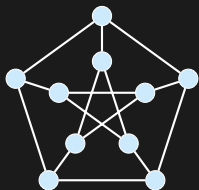


**AI is the offer made by the devil to the mind.**

The devil says: I will give you this powerful machine, it will complete any thought you like.

All you need to do is give me your soul: give up **the struggle** and you will have this marvelous machine.

— Gemini, 2025



The Petersen Graph

## Seidel Matrix $S$

$$\begin{pmatrix} 0 & - & - & 1 & 1 & - & 1 & 1 & 1 & 1 \\ - & 0 & - & - & 1 & 1 & - & 1 & 1 & 1 \\ - & - & 0 & - & - & 1 & 1 & - & 1 & 1 \\ 1 & - & - & 0 & - & 1 & 1 & 1 & - & 1 \\ 1 & 1 & - & - & 0 & 1 & 1 & 1 & 1 & - \\ - & 1 & 1 & 1 & 1 & 0 & 1 & - & 1 & - \\ 1 & - & 1 & 1 & 1 & 1 & 0 & 1 & - & - \\ 1 & 1 & - & 1 & 1 & - & 1 & 0 & - & - \\ 1 & 1 & 1 & - & 1 & 1 & - & - & 0 & - \\ 1 & 1 & 1 & 1 & - & - & - & - & - & 0 \end{pmatrix}$$

Spectrum:  $3^{(5)}, -3^{(5)}$

## Gram Matrix $G$

Construct  $X = \{1, \pm 1/3\}$ :

$$G = I + \frac{1}{3}S \succeq 0$$

---

Discovery:  $AI = \text{爱}$

- Example suggested by **Gemini**.
- Verified & Formalized by **Damase & P.**

# Soft Thresholding

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- 2 The natural measure of *faithfulness* is the maximum linear coefficient in the Gegenbauer expansion of a positive definite function vanishing on some set  $K$  to be thresholded to 0. (Closer, to one, closer to linear, and hence easier to recover the original data.)

# Soft Thresholding

- 1 Given an empirical correlation matrix of rank  $N$ , (low sample statistics) one may want to threshold small entries to 0 to obtain sparsity. However, such a process does not preserve the positive definiteness, as it is not a positive definite function.
- 2 The natural measure of *faithfulness* is the maximum linear coefficient in the Gegenbauer expansion of a positive definite function vanishing on some set  $K$  to be thresholded to 0. (Closer, to one, closer to linear, and hence easier to recover the original data.)

Theorem (Damase, P.)

*The faithfulness constant of  $[-\varepsilon, \varepsilon]$  does not go to 1 if  $N \geq 4$ .*

# Soft Thresholding

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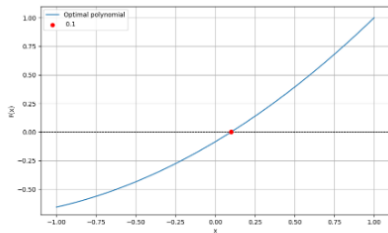


FIGURE 1. The thresholding function for  $K = \{0.1\}, n = 3$

